

Progettazione di Applicazioni Mobili

Smartphone features and their implications

Stefano Burigat

Dipartimento di Scienze Matematiche, Informatiche e
Fisiche - Università di Udine

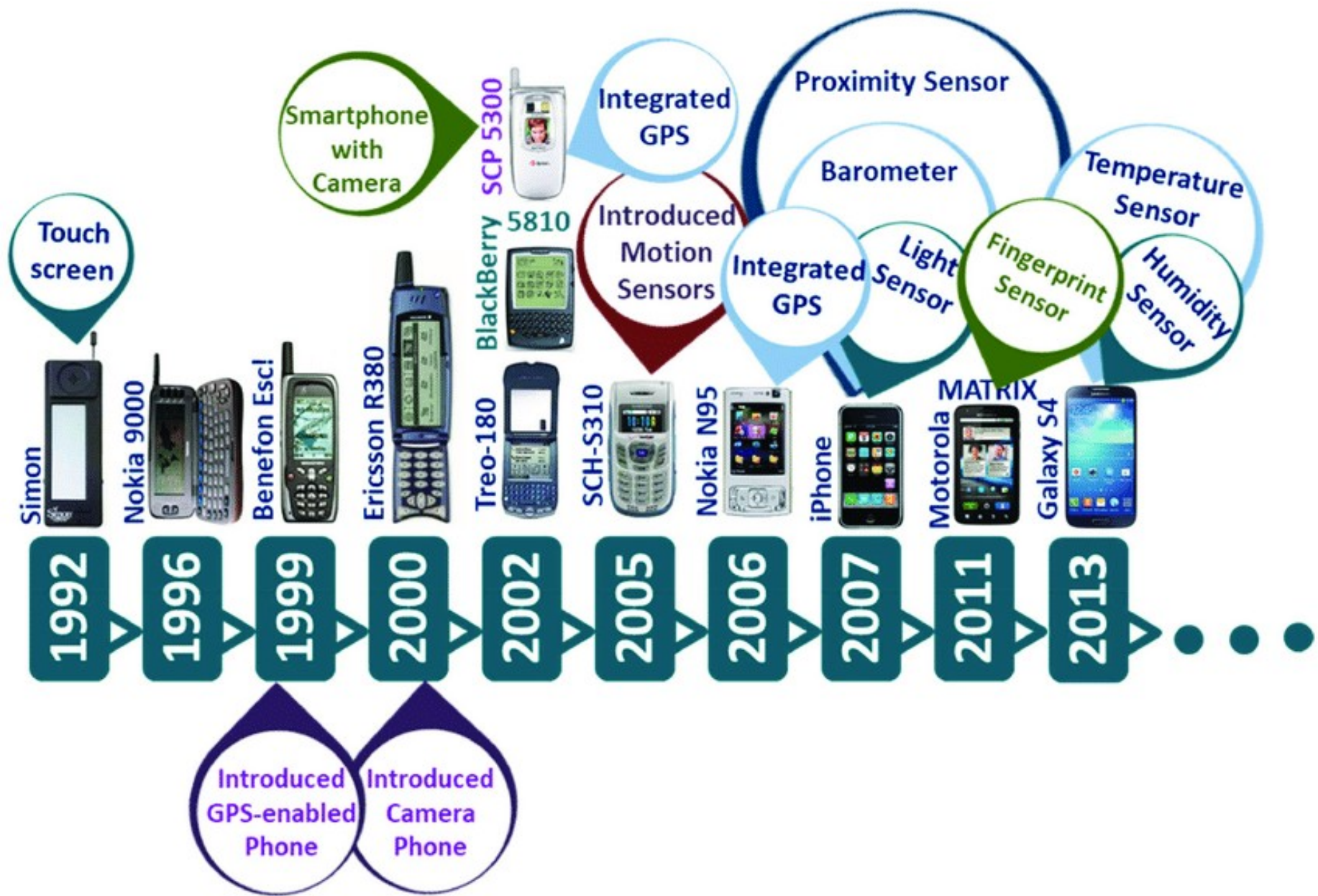
stefano.burigat@uniud.it



- Joint venture between IBM and BellSouth
- Mobile phone, pager, PDA, fax, third party applications
- Resistive touchscreen (stylus)
- DOS operating system
- Vadem 16 MHz CPU, 16-bit, x86-compatible
- 1 MB RAM
- PCMCIA memory card
- Brick form factor, 510 g

1993
The IBM Simon







15 billion

transistors

50% faster
CPU
vs. competition

2x

system cache

30% faster
GPU
vs. competition

Faster

NEURAL ENGINE



15.8 trillion

operations
per second



New video encoder



New video decoder



New ISP



New display engine

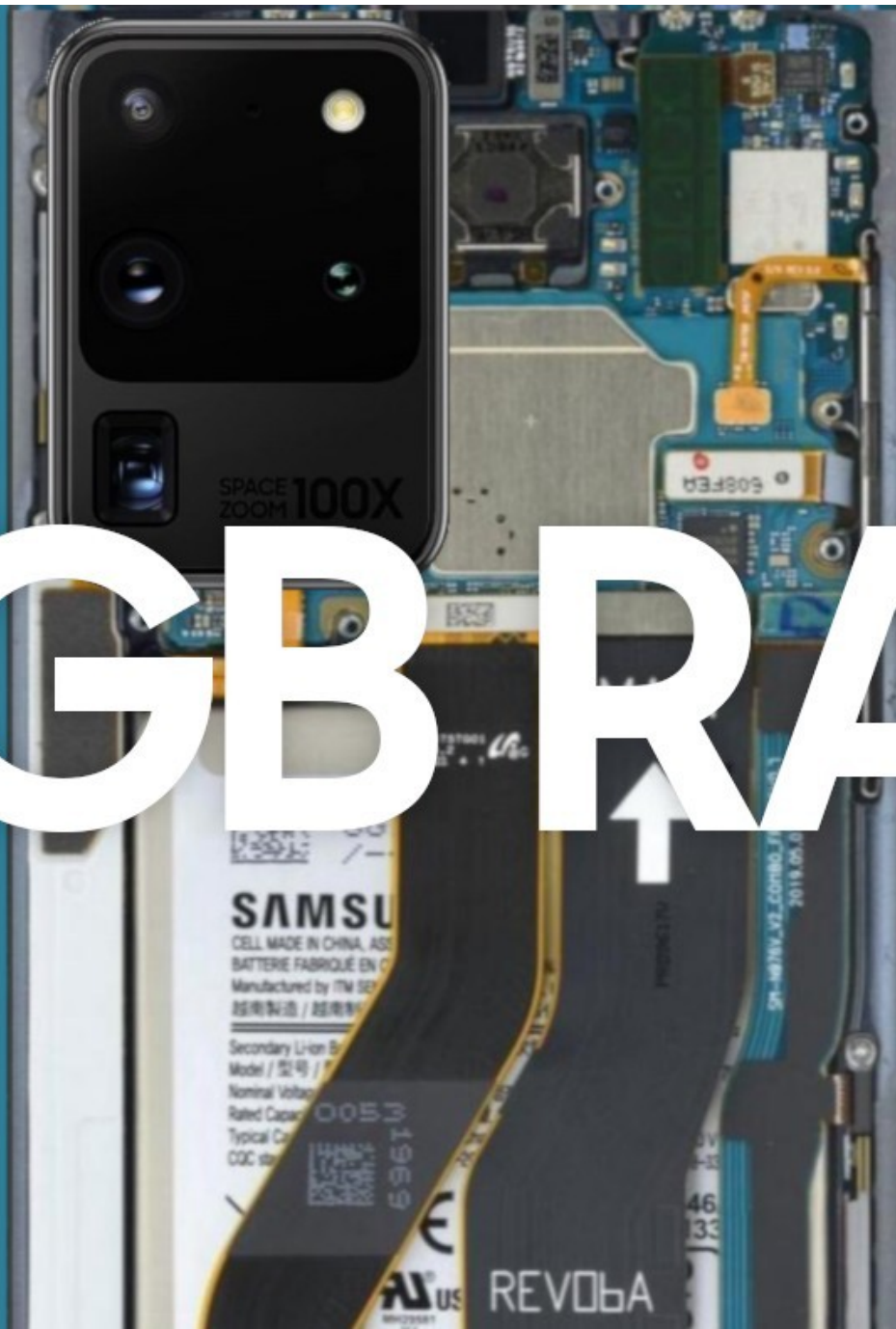


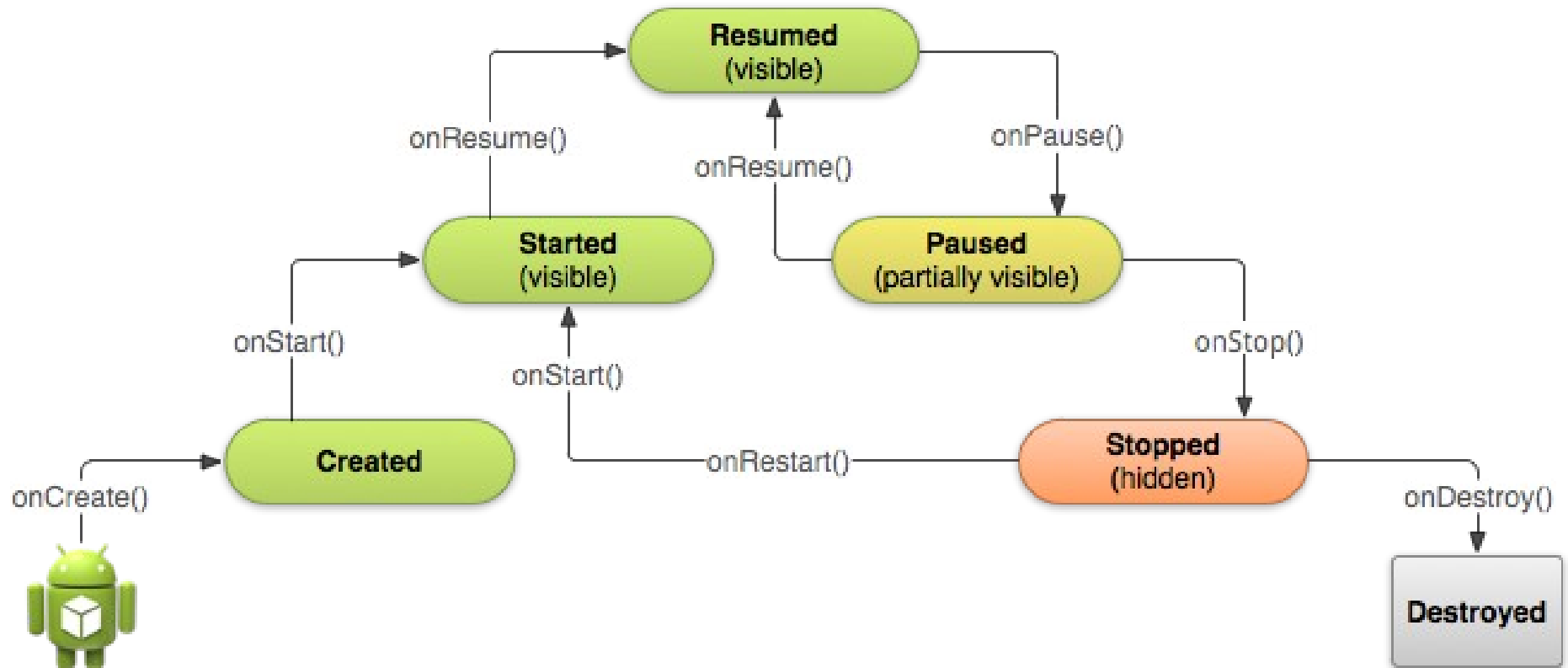
Wider lossy compression support



Secure Enclave

16GB RAM







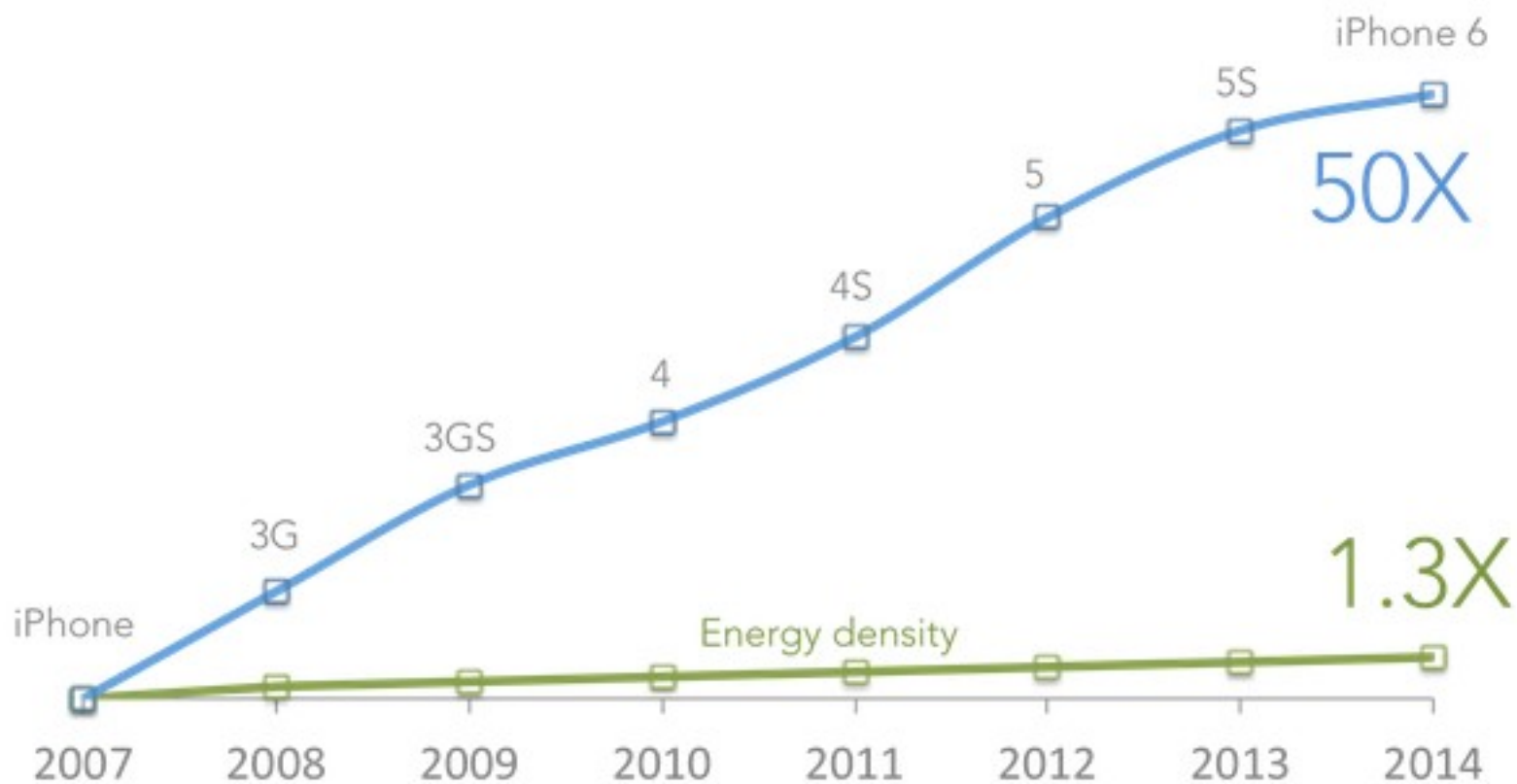
Smartphone: **Inputs** & **Outputs**

- Touch sensitive screen
- Buttons
- Microphone
- Camera
- GPS receiver
- Accelerometer
- Barometer
- Through a port
- Via a network connection



- Screen
- Speakers
- Small motor to produce vibrations
- Bright LEDs used as a flash
- Network connections





Energy density has increased by a mere 1.3X since the introduction of the first iPhone in 2007.



Sensors Everywhere

The average smartphone has at least 10 sensors.
Here are the most common.

Camera

What would you do without your selfies?

Pedometer

More and more phones are including a fitness element. Experts recommend 10,000 steps a day.

Light Sensor

Have you ever turned your phone on in the dark and had it been too bright? Your light sensor may have been malfunctioning.

Thermometer

If you've ever left your phone out in the sun you've most likely seen it turn off due to heat. The thermometer is useful to monitor internal operating temperature.

Fingerprint Sensor

The new fingerprint sensor adds an extra layer of security to your phone.

Microphone

The oldest sensor on any phone. Microphones make it possible for others to hear what you are saying.

Proximity Sensor

This is what keeps you from accidentally pressing buttons with your cheek during calls!

Magnetometer

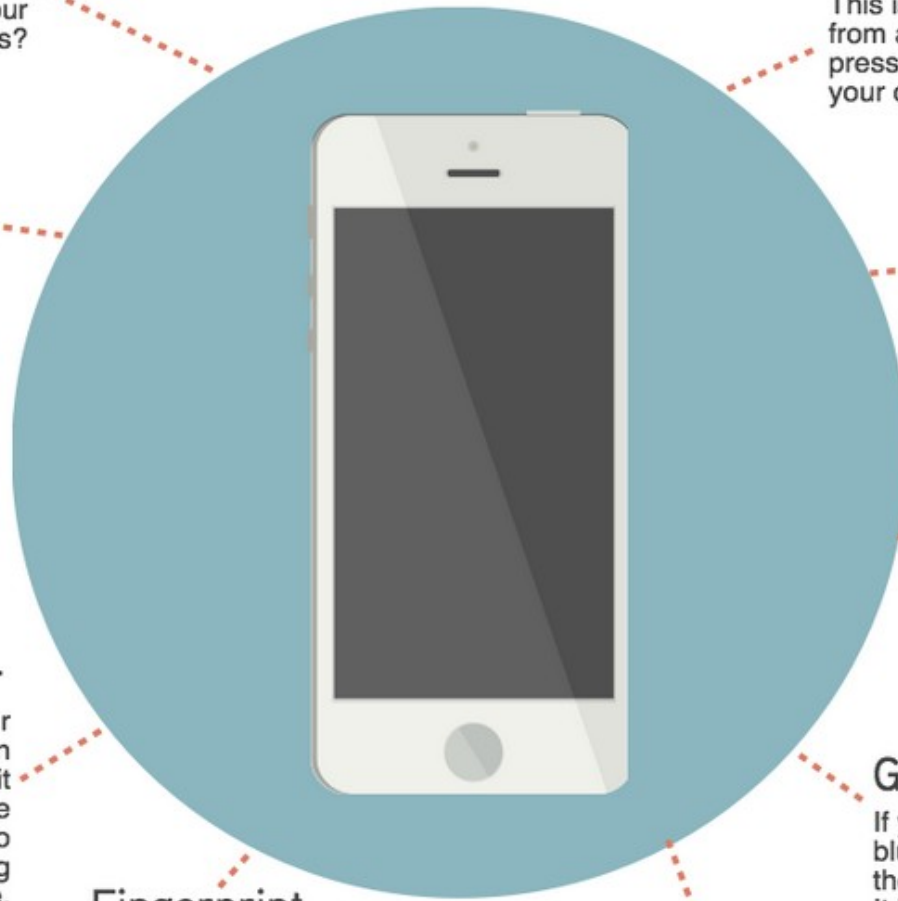
The magnetometer measures the strength of the magnetic field around the device to determine what direction it is moving.

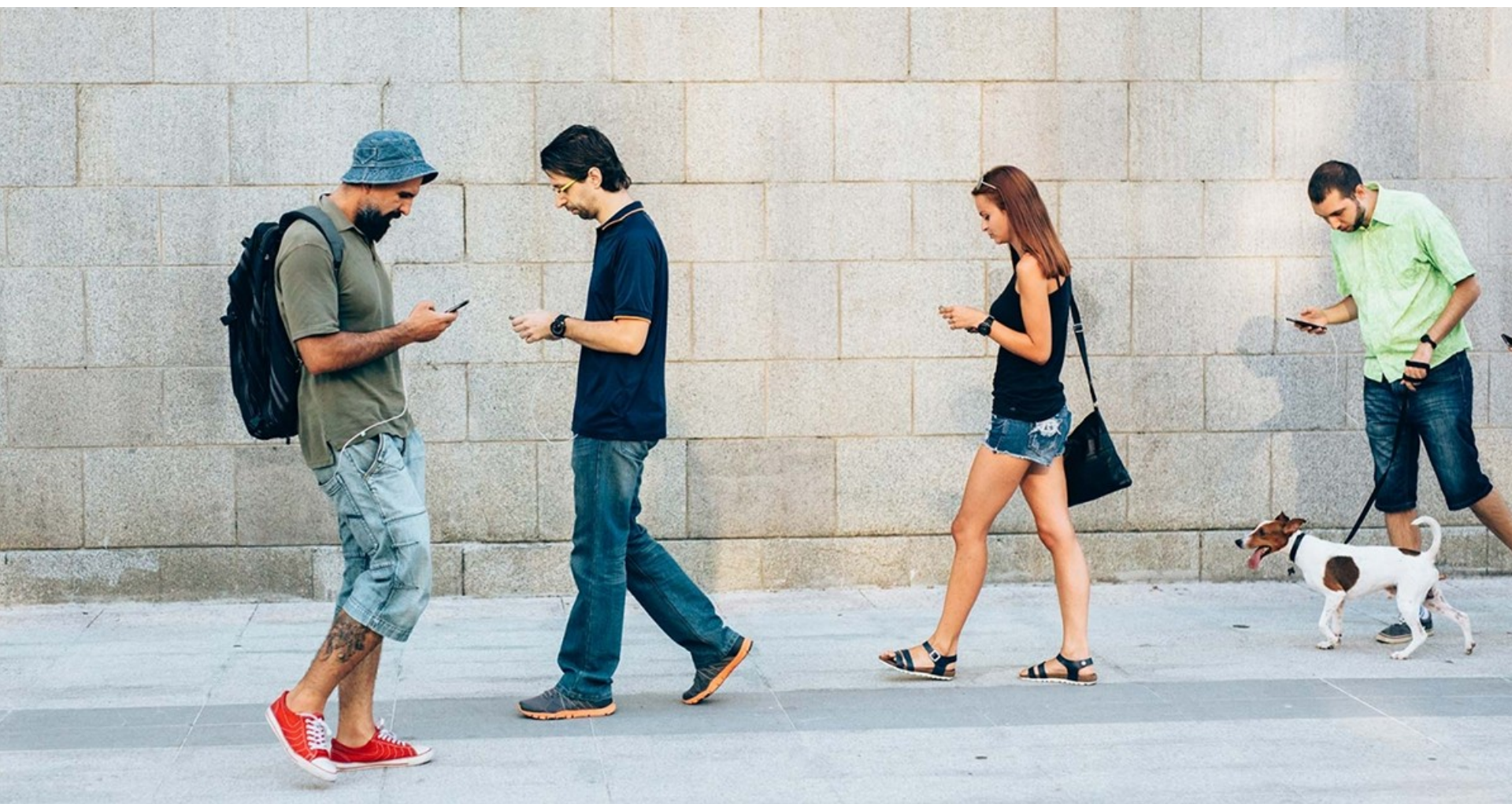
Accelerometer

Have you ever wondered how your phone knows which way you are holding it to display vertically vs. horizontally? The accelerometer is the answer!

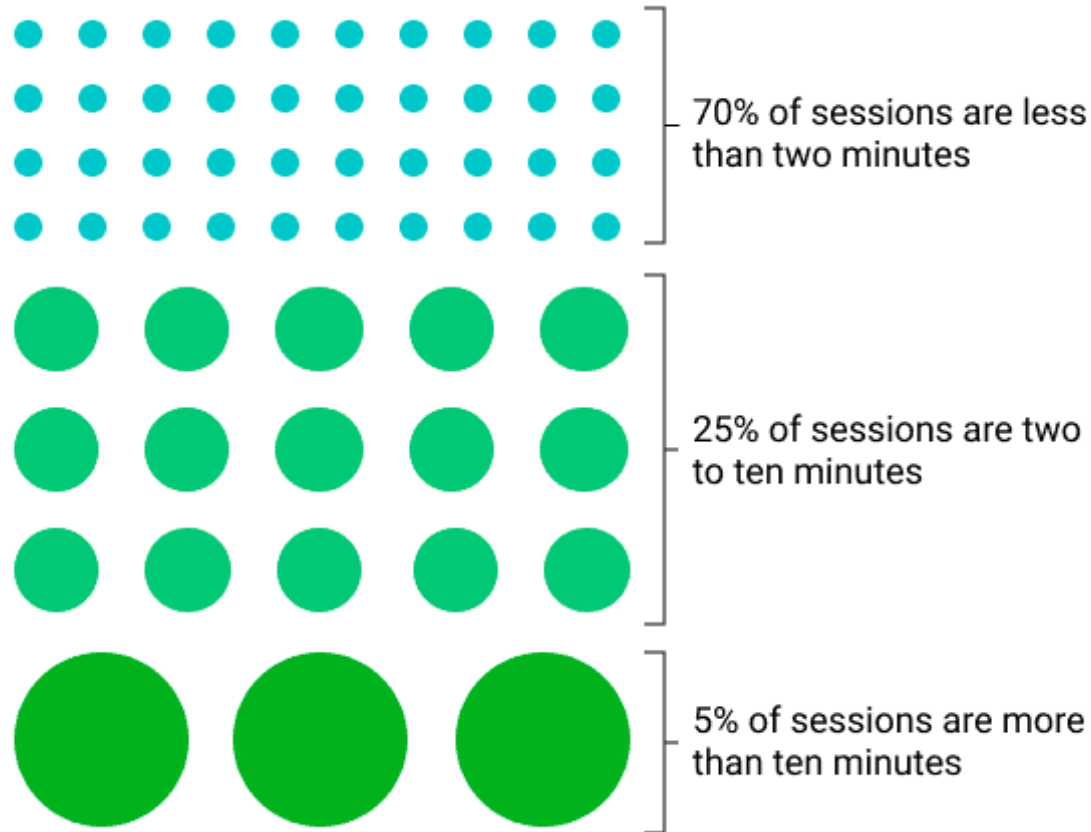
Gyroscope

If you like taking non-blurry photos you have the gyroscope to thank. It helps to correct for camera shake.



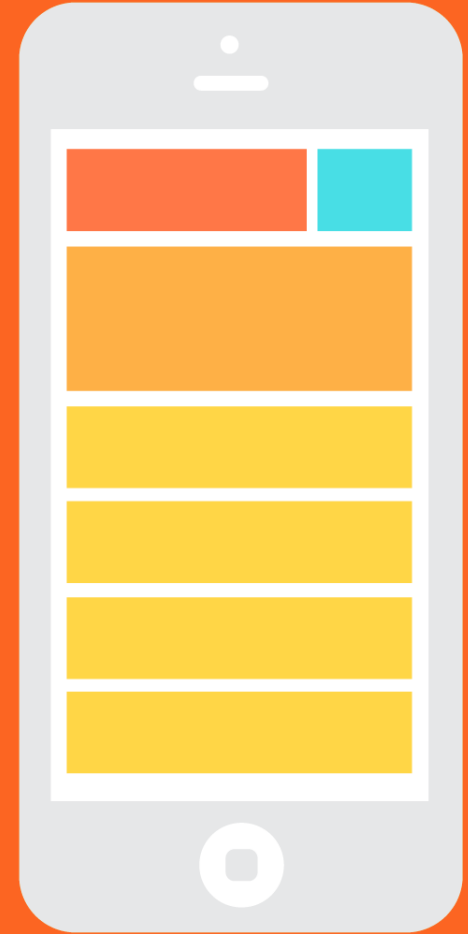
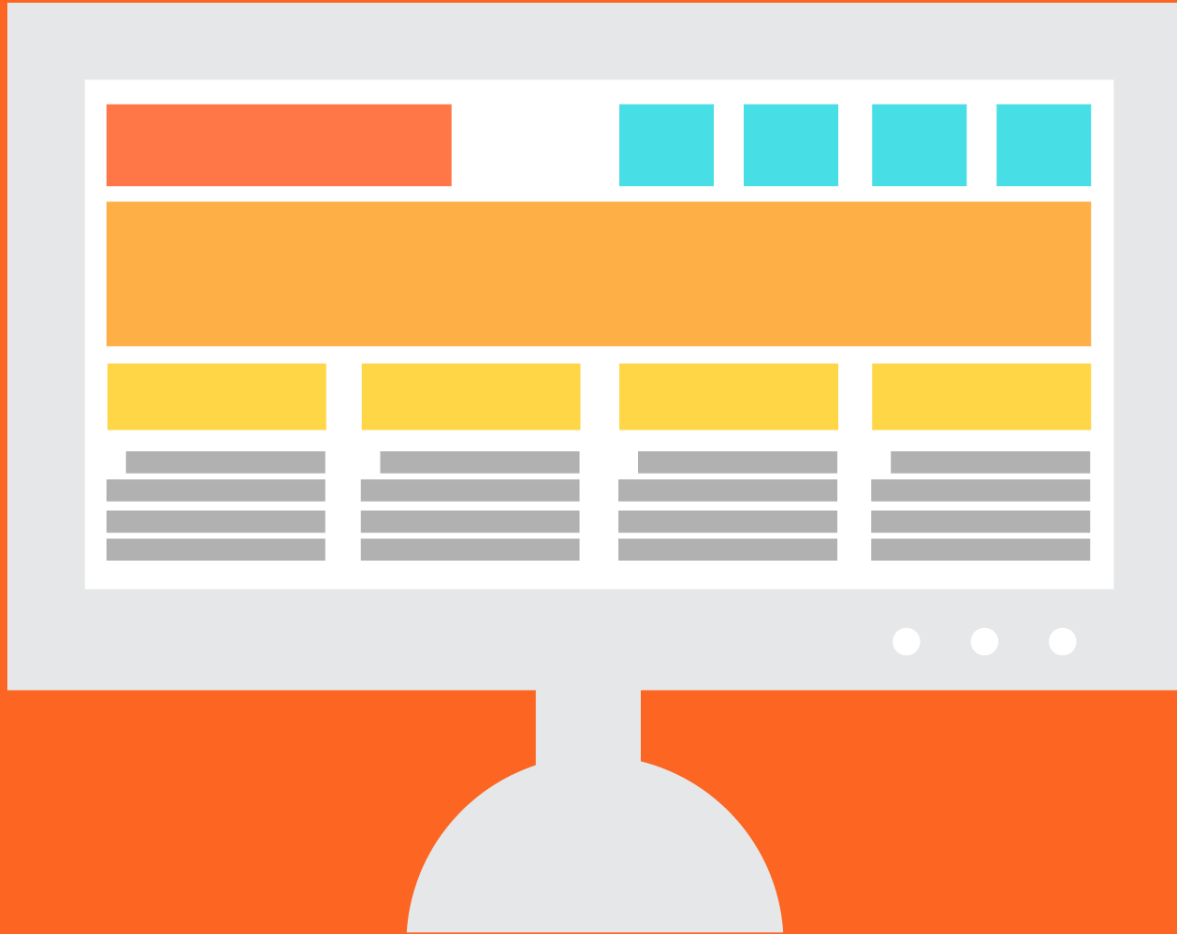


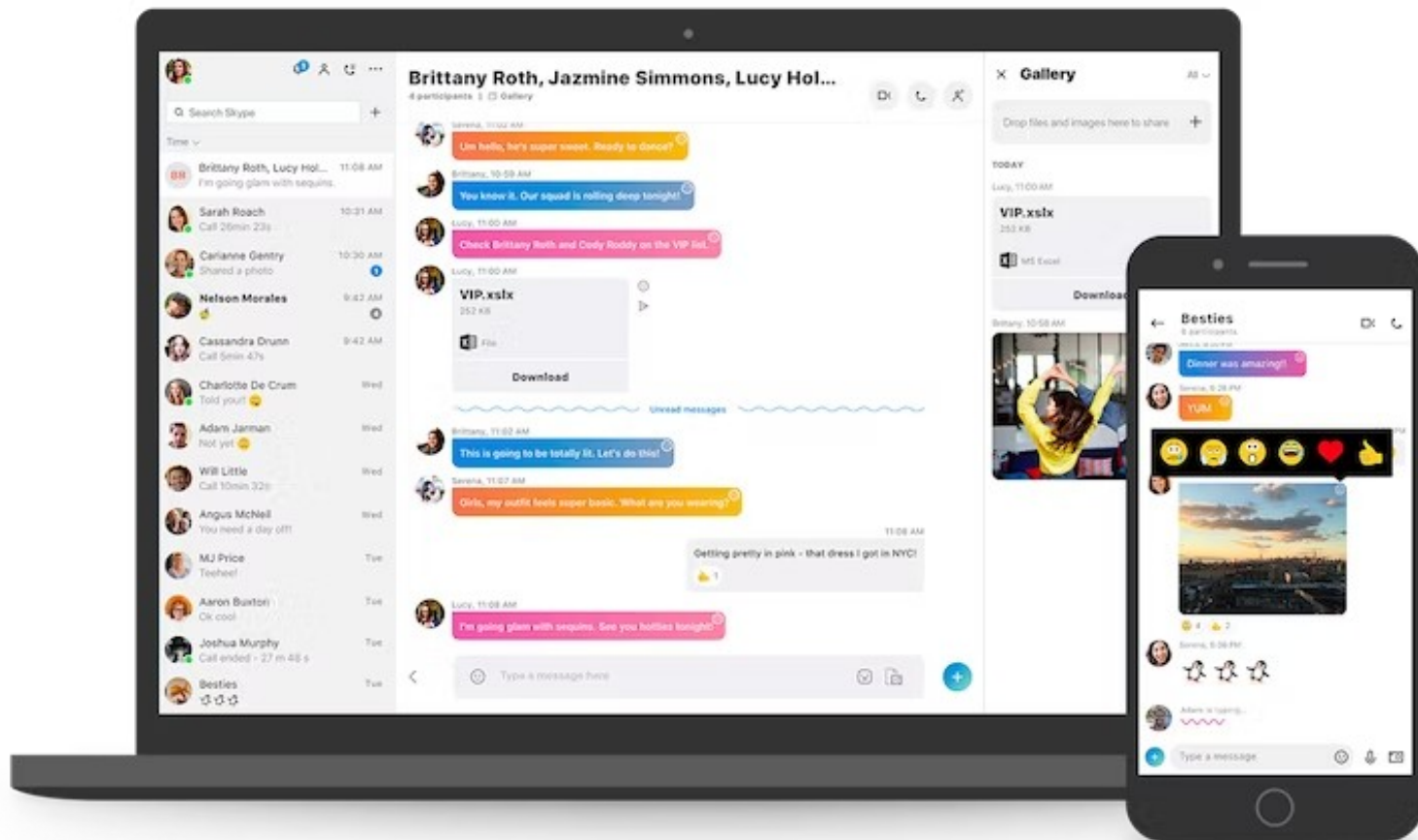
On average, we pick up our phones 58 times per day

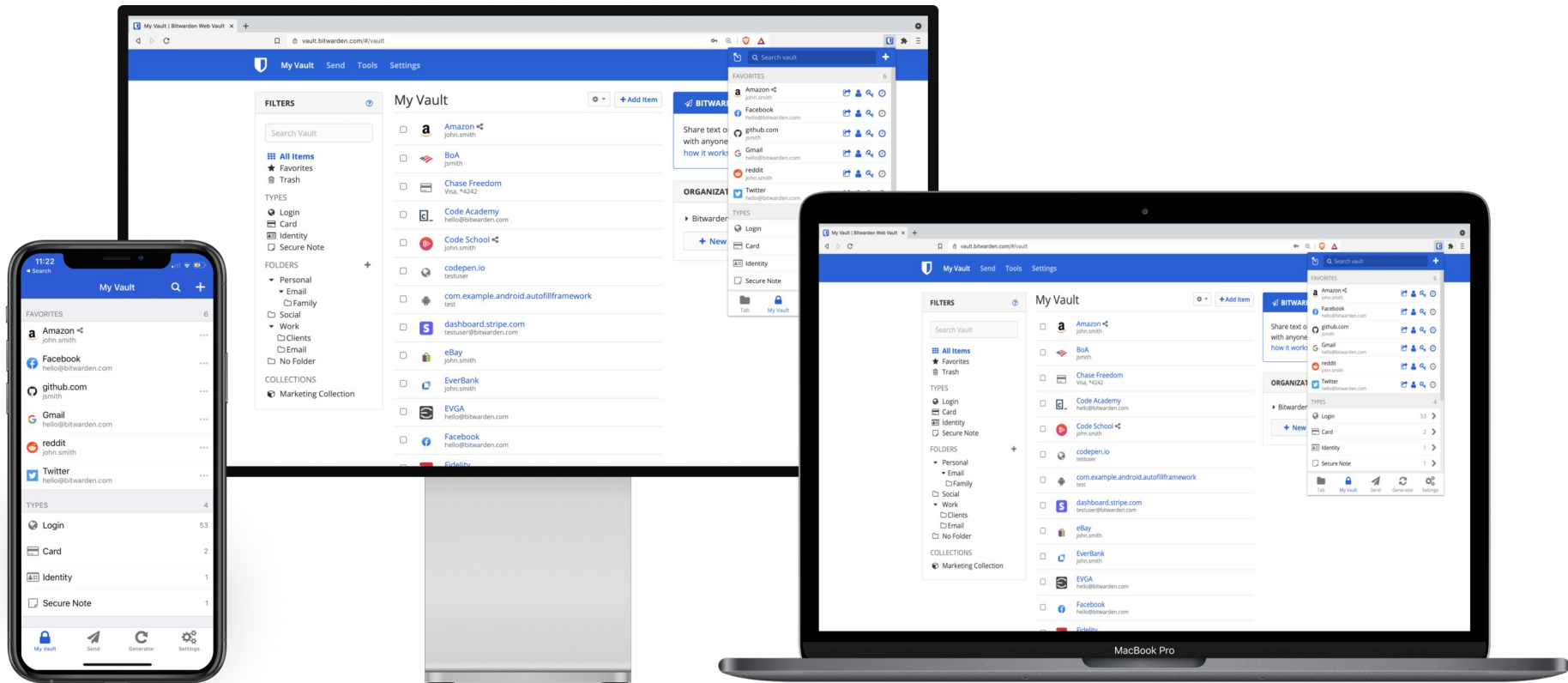


**This time adds up to about
3 hours and 15 minutes.**









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Customer Service

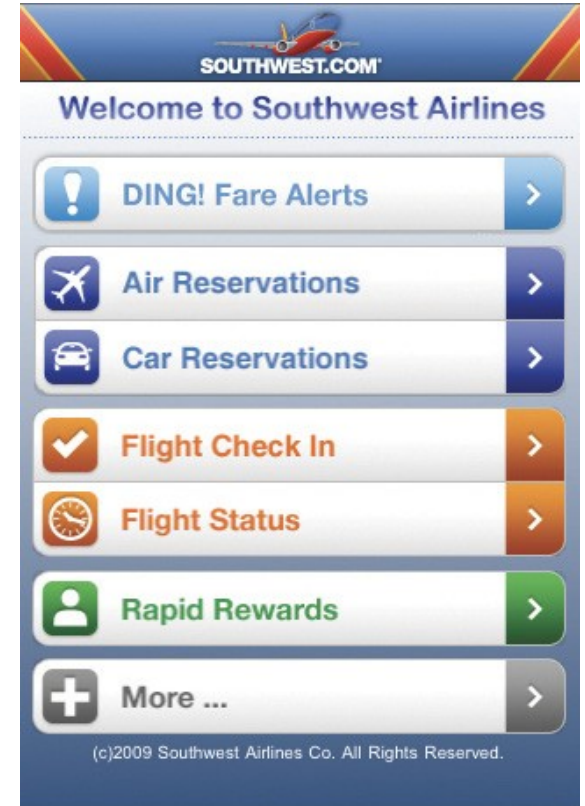
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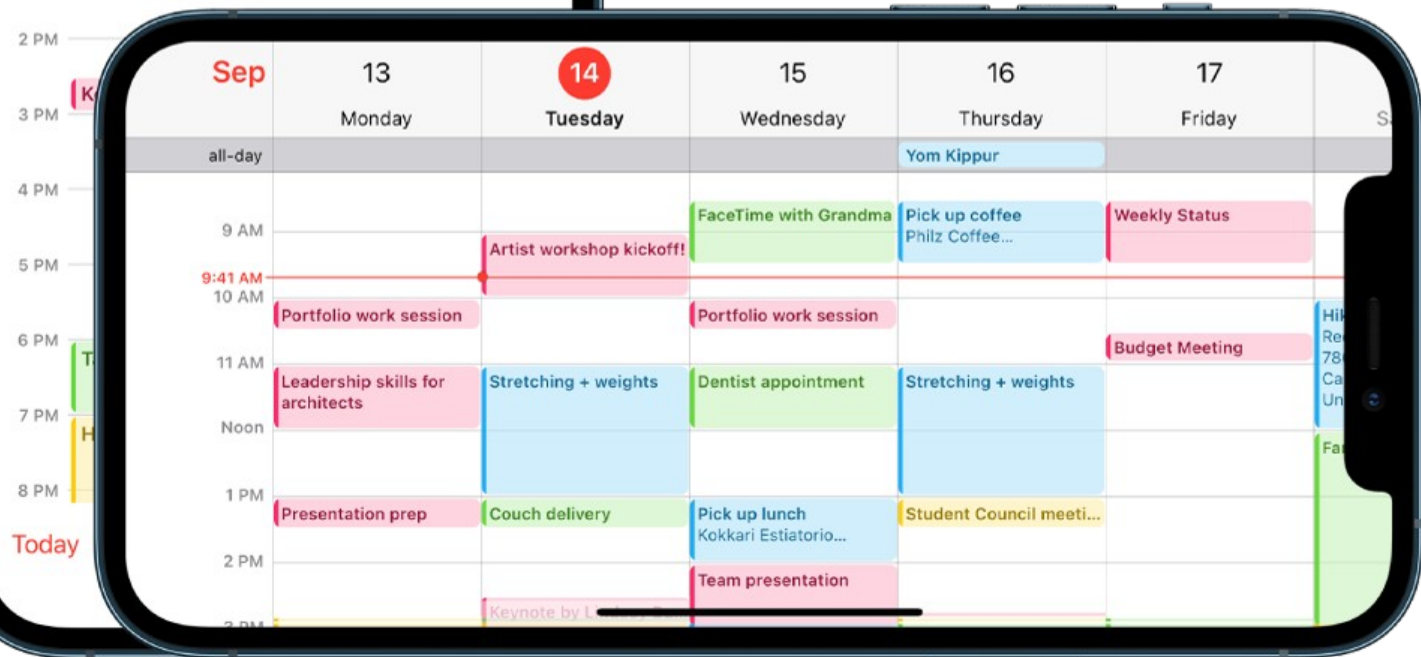
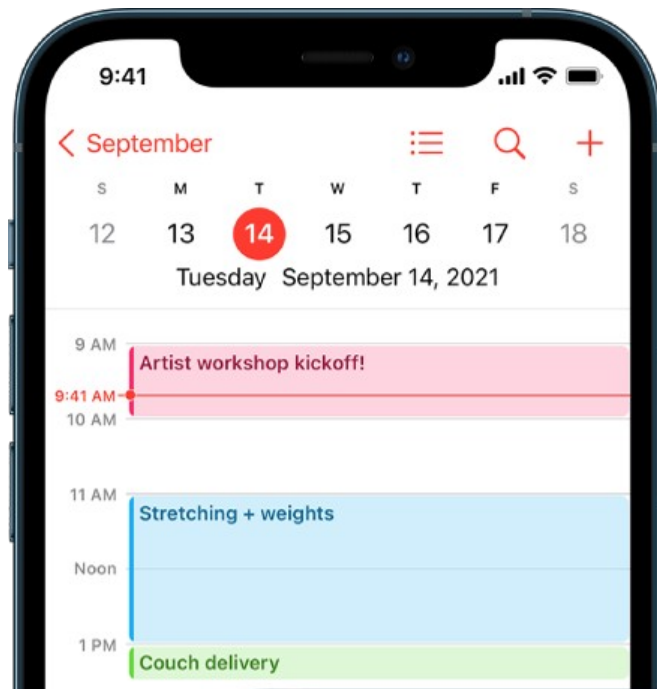


SOUTHWEST.COM

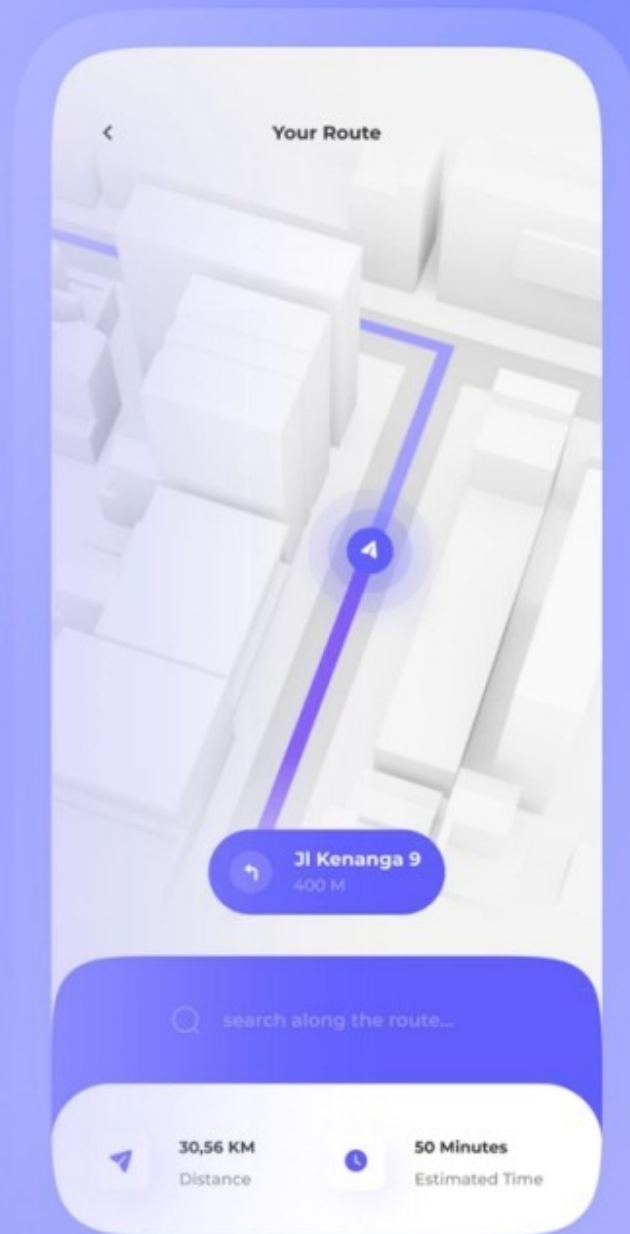
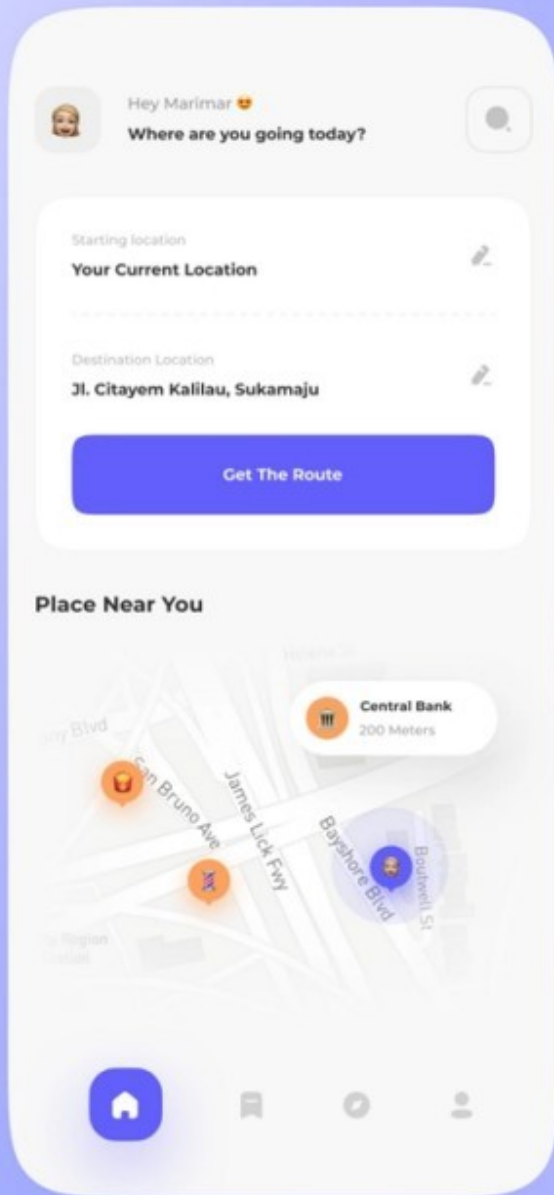
Welcome to Southwest Airlines

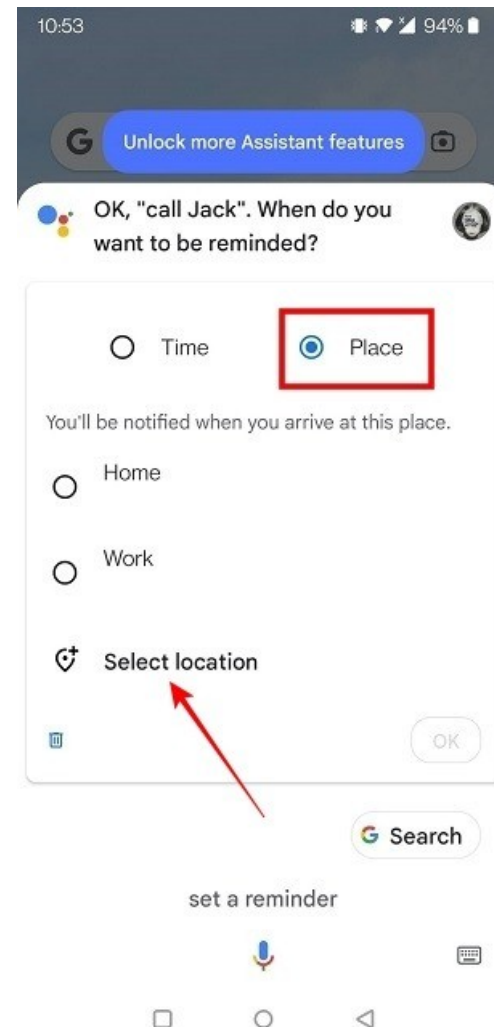
- DING! Fare Alerts**
- Air Reservations**
- Car Reservations**
- Flight Check In**
- Flight Status**
- Rapid Rewards**
- More ...**

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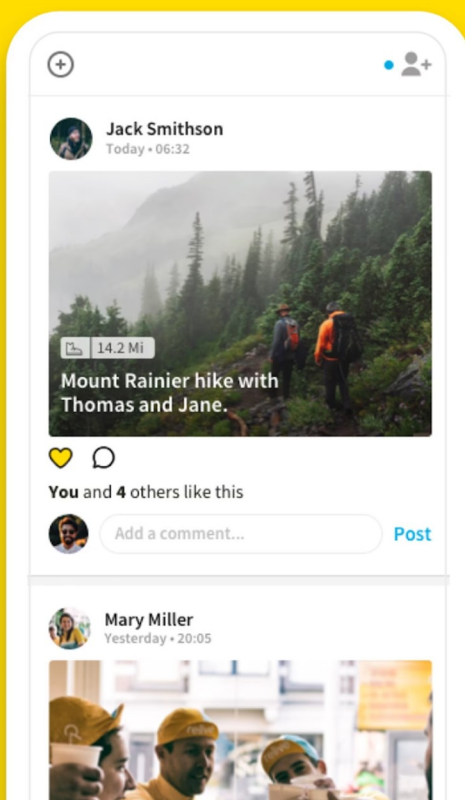




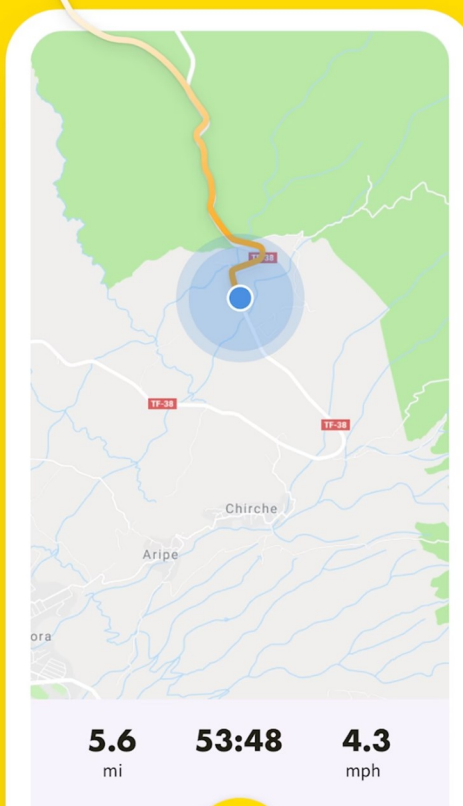




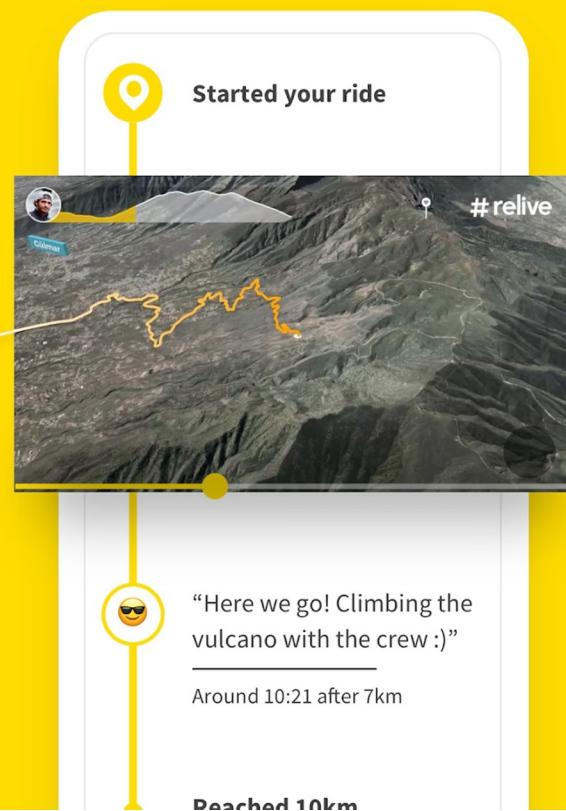
**CONTROLLA COSA FANNO I
TUOI AMICI.**



**TRACCIA LE TUE CORSE, I GIRI
IN BICI, LE PASSEGGIATE E
ALTRO.**

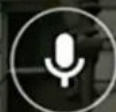


**VIDEO 3D DELLE TUE ATTIVITÀ
ALL'APERTO.**

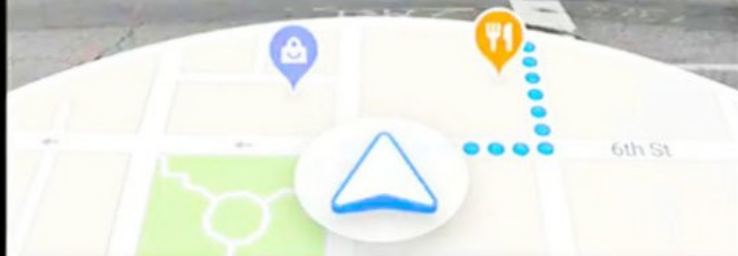




Turn right
on W 6th St



Then

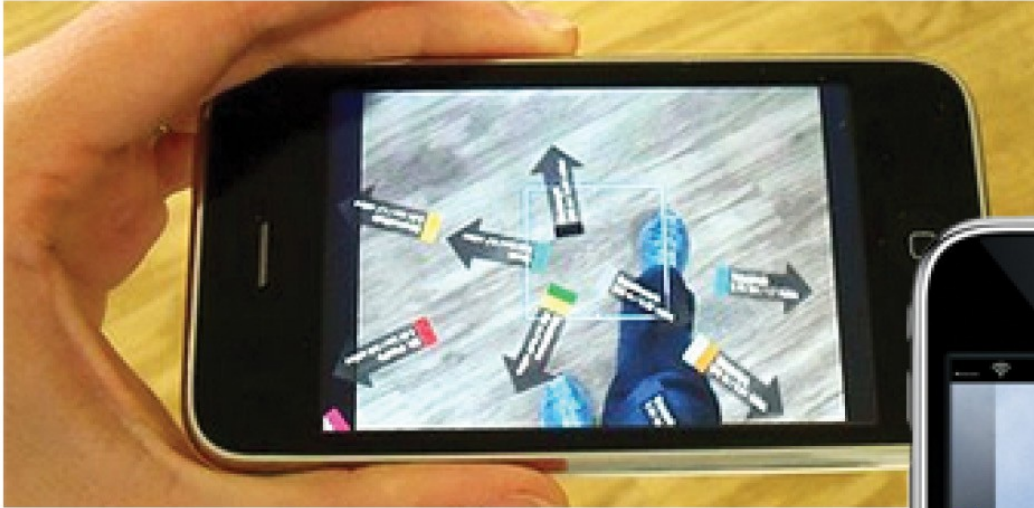


8 min

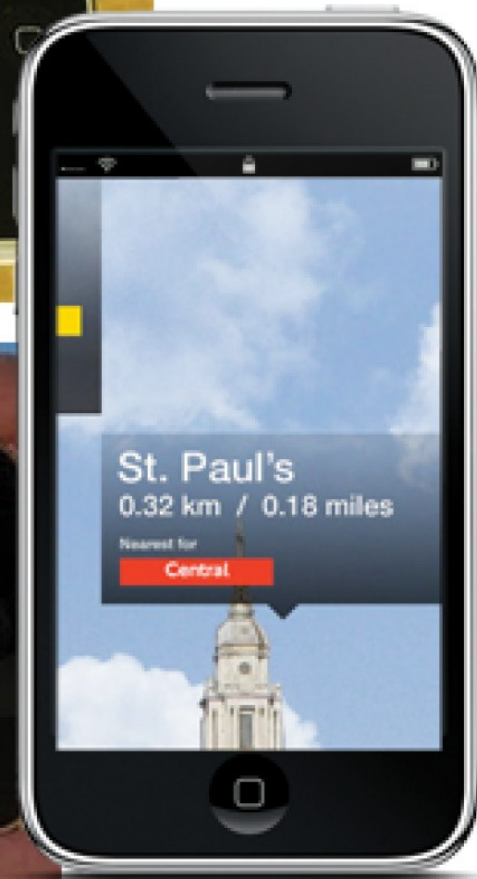
0.4 mi · 8:08 AM



Looking down



Looking ahead



Looking up

